

PS4: Does Selling Options Pay?

CHEME 4/5660 Cornell University | Mathematical Companion

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Training only. This assignment uses historical option quotes and simplified trading assumptions for teaching. It provides no investment advice.

Selling options. A short option collects its premium at entry but can lose many times that amount. A short call with no protection has no limit on its loss, and a broker can demand more money as the loss grows.

Simplifying assumptions. One contract per leg, measured per share. Every trade is held to expiration and settled at the stock's close that day, with no early exercise or assignment. We omit commissions, dividends, taxes, and interest on premiums. The quotes come from an indicative data feed.

The central question

Does selling option premium pay, and what does protection cost?

- Study 1: the promise. Build four structures from option legs. Read each one's credit and its probability of profit under four price models.
- Study 2: the outcome. Settle every trade at the stock's closing price on the expiration date. Measure its profit, its return on capital, and which model came closest to the win rate.
- Study 3: the conclusion. Decide whether selling premium beat the benchmarks and what protection cost, and see why every return is per trade.

Examples use the checks' test trade: entry price $S_0 = 100$, $D = 30$ days to expiration, and strikes (mid premiums, USD/share) of long put 80 (0.45), short put 90 (1.25), ATM 100 (put 2.90, call 3.10), short call 110 (1.05), long call 120 (0.35).

Reading guide Both tracks do Studies 1–3. The put-side and call-side split on slide 17 is Advanced only.

Building a structure from legs

A structure $\mathcal{C}$ is a set of option legs on one stock with one expiration date. Each leg i is a call or a put with strike K_i, number of contracts $n_i = 1$, premium $\mathcal{P}_i$, and orientation θ_i, all per share:

$$\theta_i = \begin{cases} -1 & \text{short: you sell leg } i\text{, and } \mathcal{P}_i \text{ is its sell price,} \\ +1 & \text{long: you buy leg } i\text{, and } \mathcal{P}_i \text{ is its buy price.} \end{cases}$$

The credit collected at entry is the premiums received minus those paid:

$$\text{credit} = -\sum_{i \in \mathcal{C}} \theta_i \, n_i \, \mathcal{P}_i .$$

In code, each leg is one call to the course `build` function, with $\texttt{K} = K_i$, $\texttt{premium} = \mathcal{P}_i$, $\texttt{sense} = \theta_i$, and $\texttt{copy} = n_i = 1$.

Also set $\texttt{DTE} = D/365$, the years to expiration, and `IV`, the leg's implied volatility. The contract types require them; neither changes the profit at expiration.

The eight structures

Each trade offers six legs. Their strikes sit around the entry price S_0, in this order:

$$K_{\text{long put}} < K_{\text{short put}} < K_{\text{ATM}} \approx S_0 < K_{\text{short call}} < K_{\text{long call}}.$$

The at-the-money (ATM) call and put share the strike nearest S_0. The short call and put are near 0.25 delta in absolute value, and the long call and put, the wings, near 0.10.

Structure	You sell ($\theta_i = -1$)	You buy ($\theta_i = +1$)
Short straddle	ATM call, ATM put	nothing
Short strangle	short call, short put	nothing
Protected strangle	short call, short put	long call
Iron condor	short call, short put	long call, long put
Short put (Advanced)	short put	nothing
Short call (Advanced)	short call	nothing
Put credit spread (Advanced)	short put	long put
Call credit spread (Advanced)	short call	long call

Profit at expiration

At expiration, a call and a put with strike K pay, per share:

$$V_c(K, S(T)) = \left(S(T) - K\right)^+, \qquad V_p(K, S(T)) = \left(K - S(T)\right)^+.$$

Here $S(T)$ is the closing price on the expiration date and $(x)^+ = \max(x, 0)$. With V_i the payoff of leg i, V_c or V_p, the structure's profit is:

$$\hat{P}_{\mathcal{C}}(S(T)) = \sum_{i \in \mathcal{C}} \theta_i \, n_i \left(\underbrace{V_i(K_i, S(T))}_{\text{payoff}} - \underbrace{\mathcal{P}_i}_{\text{premium}} \right).$$

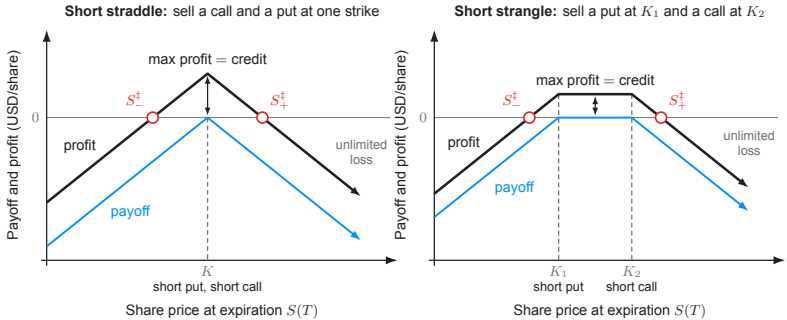
The course `profit` function returns one row per share price: the price, one column per leg, and the total $\hat{P}_{\mathcal{C}}$ in the last column.

Graded runs price every leg at its mid, $(\text{bid} + \text{ask})/2$; the *Fill prices and dollars* table compares bid/ask fills.

No interest PS4 omits `profit`'s optional accumulation factor $\mathcal{D}$ (default 1), which would grow premiums.

Profit diagrams: short legs only

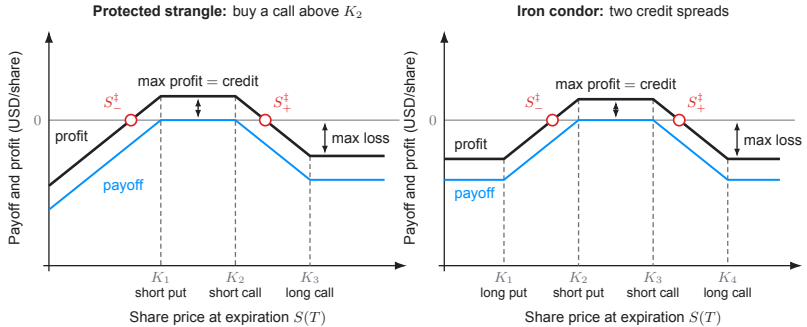
The profit (black) is the structure's payoff $\sum_i \theta_i n_i V_i$ (blue) moved up by the credit.



Test trade Mid prices, USD/share; $K_1 < K_2 < \cdots$ label each diagram's own strikes. Straddle: credit 6.00, breakevens (zero profit) 94 and 106. Strangle: credit 2.30, breakevens 87.70 and 112.30.

Profit diagrams: adding long legs

Each long leg lowers the credit and caps the loss on its side.



Test trade Protected strangle: credit 1.95, breakevens 88.05 and 111.95, loss capped at 8.05 above 120 but not on a fall. Iron condor: credit 1.50, breakevens 88.50 and 111.50, loss capped at 8.50.

Return on capital and the margin rule

Let C be the capital set aside at entry, in USD/share. The return on capital is the profit at expiration divided by the capital, $R = \hat{P}_{\mathcal{C}}(S(T))/C$, a fraction for one trade, not the lectures' annualized growth rate g.

A naked short leg (no protection) with strike K and premium $\mathcal{P}$ needs the capital below. Here S_0 is the entry price, $m = 0.20$ the margin rate, and $f = 0.10$ the floor:

$$C_{\text{call}} = \mathcal{P} + \max(mS_0 - (K - S_0)^+,\ fS_0), \qquad C_{\text{put}} = \mathcal{P} + \max(mS_0 - (S_0 - K)^+,\ fK).$$

- Straddle, strangle: $\max(C_{\text{call}}, C_{\text{put}})$; only one side can lose.
- Credit spread: the strike width minus its own credit, its largest loss.
- Iron condor: the wider of its two widths minus the total credit.
- Protected strangle: the larger of C_{put} and the call credit spread's largest loss.

Test trade At mid prices, C is 23.10 (straddle), 11.25 (short and protected strangles), and 8.50 (iron condor) USD/share.

Return on capital versus the share price

Divide the profit and the capital by the entry price S_0:

$$R = \frac{\hat{P}_{\mathcal{C}}(S(T))/S_0}{C/S_0} = \frac{\text{profit as a fraction of } S_0}{\text{capital as a fraction of } S_0}.$$

The report's *Capital (% of share price)* column averages C/S_0. The test trade's strangle sets aside 11.25% of S_0, so a profit or loss of 1 USD/share, 1% of S_0, is about 8.9% of capital.

- Every short leg protected. A credit spread's lowest profit is its credit minus its width, which is $-C$. The same holds for an iron condor, so $R \geq -100\%$.
- A short leg without protection. A short call can lose without limit, and a short put up to its strike minus its premium. Then $R < -100\%$ is possible, and a broker would demand more money. The protected strangle's put is unprotected.

Probability of profit by counting

Each model on the next slide supplies $N = 10{,}000$ simulated prices at expiration, $S^{(1)}(T), \ldots, S^{(N)}(T)$. With the indicator $\mathbf{1}[\,\cdot\,]$, one when its condition holds and zero otherwise, the probability of profit is the fraction of prices with a positive total profit:

$$\widehat{\mathrm{POP}} = \frac{1}{N} \sum_{j=1}^{N} \mathbf{1}\Big[\hat{P}_{\mathcal{C}}(S^{(j)}(T)) > 0\Big].$$

A simulated price with exactly zero profit does not count. Counting needs no breakeven formula, so it works for any structure.

A trade is won when its profit at the one actual closing price is positive, $\hat{P}_{\mathcal{C}}(S(T)) > 0$.

Precision With $N = 10{,}000$, the standard error $\sqrt{\widehat{\mathrm{POP}}(1 - \widehat{\mathrm{POP}})/N}$ is at most 0.5 percentage points per trade. A win rate over 174 trades has a standard error of up to 3.8 points.

The four GBM models

Each model is a geometric Brownian motion with drift μ and volatility σ per year. The course `sample` function steps one day, $\Delta t = 1/365$ year, from entry at price S_0 to expiration at $T = D/365$:

$$S_{t_j} = S_{t_{j-1}} \exp\Big[(\mu - \tfrac{\sigma^2}{2})\Delta t + \sigma\sqrt{\Delta t}\, Z_j\Big], \qquad t_j = j\,\Delta t, \quad Z_j \sim \mathcal{N}(0,1).$$

The models differ only in μ and σ. The risk-free rate is $r = 0.0376$ per year, and σ_{IV} is the mean of the ATM call's and put's implied volatilities at entry:

Report column	Drift μ	Volatility σ
Model POP: market (r, IV)	r	σ_{IV}
Model POP: historical drift, IV	$\hat{\mu}$ (slide 13)	σ_{IV}
Model POP: r, historical vol	r	$\hat{\sigma}$ (slide 13)
Model POP: historical drift and vol	$\hat{\mu}$	$\hat{\sigma}$

Same draws The four models of a trade reuse the same Z_j, so their probabilities differ because of μ and σ, not chance.

Historical drift and volatility

From the last 253 daily closes before the entry date, $c_0, \ldots, c_{252}$, with log changes $\ell_k = \ln(c_k/c_{k-1})$ and their mean $\bar{\ell}$, the annualized log growth rate $\hat{g}$ and volatility $\hat{\sigma}$ are:

$$\hat{g} = 252\,\bar{\ell}, \qquad \hat{\sigma} = \sqrt{252}\,\sqrt{\frac{1}{251}\sum_{k=1}^{252}(\ell_k - \bar{\ell})^2}.$$

The course `sample` grows the log price at $\mu - \sigma^2/2$, so the historical drift that gives log growth $\hat{g}$ is:

$$\hat{\mu} = \hat{g} + \frac{\hat{\sigma}^2}{2}, \qquad \text{so that} \qquad \hat{\mu} - \frac{\hat{\sigma}^2}{2} = \hat{g}.$$

The historical drift, IV model changes only the market model's drift, to $\hat{\mu}$; its log price grows at $\hat{\mu} - \sigma_{\mathrm{IV}}^2/2$, equal to $\hat{g}$ only if $\sigma_{\mathrm{IV}} = \hat{\sigma}$.

Two calendars Daily changes are annualized with 252 trading days, T with 365 calendar days. They agree: about 21 trading days span 30 calendar days.

Win rate, average return, and probability of profit

For the n trades of one structure, let P_k be trade k's profit at expiration, R_k its return on capital, and POP_k a model's probability of profit. Three averages answer three questions:

$$\underbrace{W = \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}[P_k > 0]}_{\text{how often it won}}, \qquad \underbrace{\bar{R} = \frac{1}{n}\sum_{k=1}^{n} R_k}_{\text{what it earned}}, \qquad \underbrace{\overline{\mathrm{POP}} = \frac{1}{n}\sum_{k=1}^{n} \mathrm{POP}_k}_{\text{what a model promised}}.$$

Split the trades into wins and the others, with average returns $\bar{R}_{\text{win}}$ and $\bar{R}_{\text{loss}} \le 0$:

$$\bar{R} = W\,\bar{R}_{\text{win}} + (1 - W)\,\bar{R}_{\text{loss}}.$$

So $\bar{R} < 0$ whenever $(1 - W)\,|\bar{R}_{\text{loss}}| > W\,\bar{R}_{\text{win}}$, however high W is. A few large losses move the mean far more than the median, which the report also lists.

Equal weights $\bar{R}$ is the return of putting the same dollar capital into every trade.

Exposure per 1% stock move

A leg's delta Δ_i, from the entry quotes, is its price change per $+1$ USD/share rise in the share price. The structure's delta adds the legs:

$$\Delta_{\mathcal{C}} = \sum_{i \in \mathcal{C}} \theta_i \, n_i \, \Delta_i .$$

A 1% rise moves the price by $0.01\,S_0$ and the structure's value by about $\Delta_{\mathcal{C}} \times 0.01\,S_0$. The report's *Change in value per 1% stock rise* divides this by the capital:

$$E = \frac{\Delta_{\mathcal{C}} \times 0.01\,S_0}{C}.$$

A share bought outright has $E = 1\%$; $E = 2\%$ is twofold leverage. The test trade's short put ($\Delta_i = -0.25$, $\theta_i = -1$) has $E = 0.25 \times 1.00/11.25 \approx 2.2\%$.

First order An $x\%$ move changes the value by about $x\,E$ of capital. Deltas change as the stock moves: a short straddle starts near $E = 0$ yet loses on a large move either way.

Why annualizing a per-trade return misleads

A trade lasts $D \approx 30$ calendar days, so a year holds about $365/D \approx 12$ trades. Two common ways to turn a per-trade return R into a yearly figure are:

$$\underbrace{\frac{365}{D}\,R}_{\text{simple}} \qquad \text{and} \qquad \underbrace{(1+R)^{365/D}-1}_{\text{compounded}}.$$

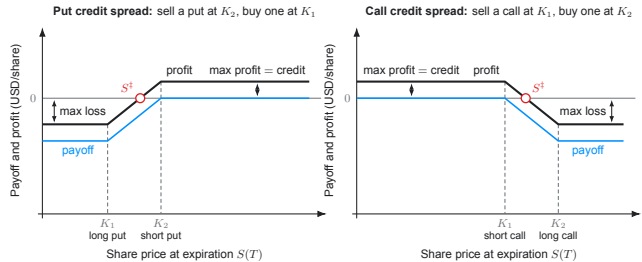
Both assume every month earns the same R, which causes three problems:

- the next twelve trades need not win and lose as the sampled trades did;
- an average over a limited sample with rare large losses is uncertain, and multiplying it by 12 multiplies that uncertainty by 12;
- compounding reinvests all capital, which needs $1 + R > 0$ every month.

The test trade's strangle earns $R = +20.4\%$ at an unchanged price; multiplied by 12, about one trade a month, that gives about $+245\%$. A close at 130 USD/share gives $R = -157.3\%$, so $1 + R < 0$ and compounding breaks down.

Advanced: the put side and the call side

The iron condor is the two credit spreads below. Their profits add, $\hat{P}_{\mathcal{C}}(S(T)) = \hat{P}_{\text{put side}}(S(T)) + \hat{P}_{\text{call side}}(S(T))$, but their returns on capital do not: each side has its own capital.



Also split The short strangle (short put, short call) and the protected strangle (short put, call credit spread). Compare the sides by average profit.

Summary

PS4 builds option-selling structures from legs, compares what price models promised with what happened at expiration, and measures each profit against the capital a broker would require.

Key takeaways

- A structure's profit at expiration is the sum of its legs' profits. A long leg costs part of the credit and caps the loss on its side.
- The model probability of profit, the win rate, and the average return on capital answer different questions. A high win rate alone does not give a positive average.
- Return on capital and exposure are measured against the capital, a fraction of the share price, so they swing far more than the stock's own return.

Use these measures, per trade and never per year, to answer the central question.